

Assessing ChatGPT's Efficacy in Interpreting Privacy Policies

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Abstract: In the contemporary digital landscape, the intricacy of privacy policies poses a considerable challenge, often leaving users confused and at risk of compromising their privacy. This position paper delves into the capabilities of generative AI, with a particular focus on ChatGPT-4, to aid users in deciphering, summarizing, and interpreting an array of 102 privacy policies. Utilizing the Flesch-Caid readability score as a benchmark, our findings reveal that ChatGPT enhanced the readability of 56 out of the 102 policies evaluated. However, in certain instances, the interpretations by ChatGPT resulted in reduced readability (such as—), highlighting potential limitations of the current state of AI in processing complex legal text. Despite this, the application of ChatGPT's capabilities is seen as a pivotal step towards distilling complex legal verbiage into explanations that are accessible with lesser complex jargon, thereby enabling users to make well-informed decisions.

Motivation: Privacy policies are legal documents that outline how organizations collect, use, and protect personal data. Despite their importance in the digital age, the complexity and length of these documents often pose a significant barrier to user understanding and awareness [2, 3]. The rise of generative AI, and particularly LLMs like ChatGPT, presents a novel opportunity to address this challenge. By leveraging the capabilities of LLMs, there is potential to transform these dense legal texts into user-friendly explanations and summaries. Such AI-mediated interpretations can empower users to make informed decisions regarding their privacy by enhancing transparency and accessibility of policy information [4]. In the realm of data privacy, where user trust is paramount, the utilization of LLMs could mark a significant step towards demystifying the often opaque practices of data handling by organizations.

Method: To assess the potential impact of ChatGPT-4 on the readability of privacy policies across various industries, our research team meticulously compiled a robust dataset consisting of 102 privacy policies referenced from our preceding investigation [1]. These policies spanned an extensive array of sectors, encompassing food, technology, finance, and more, to ensure comprehensive industry representation. In alignment with ChatGPT's processing capabilities during our evaluation period from June 2023 to December 2023, we segmented each policy document into portions not exceeding 25,000 words. For each segment, we crafted specific prompts to ensure consistency and clarity in the task presented to ChatGPT, thereby enabling the model to maintain contextual integrity and generate cohesive summaries. After processing, we obtained AI-derived interpretations of the privacy policies, which we then subjected to a rigorous comparative analysis using the 'py-readability-metrics' Python library. This analysis involved calculating the Flesch Reading Ease scores for both the original texts and their AI-generated counterparts. Further to the readability assessment, we performed a suite of statistical tests, including the paired samples t-test, to quantitatively evaluate the differences in readability between the original and summarized documents. Further details pertaining to our methods, including the prompts used and the scripts for readability evaluation are available in the supplementary materials.

Results: We analyzed 102 privacy policies and evaluated Flesch readability score ranging from 1-100, the qualitative grade level from elementary to beyond college level, and perceived difficulty from easy to difficult. From Figure 3 we can see that ChatGPT improved 56% of all privacy policies' readability scores. In examining the comparative readability of original corporate texts versus their ChatGPT interpretations, a paired samples t-test was also conducted to determine whether there

was a statistically significant difference in readability scores. The analysis included a sample of N paired scores, where each pair consisted of the readability score for the original text and the ChatGPT interpretation corresponding to a single company. The results indicated that the mean readability score for original texts ($M_{\text{original}} = 42.5, SD_{\text{original}} = 5.2$) was significantly higher than the mean score for ChatGPT interpretations ($M_{\text{ChatGPT}} = 35.3, SD_{\text{ChatGPT}} = 6.4$), $t(N - 1) = 10.52$, $p < .001$, with a 95% confidence interval for the mean difference ranging from 6.4 to 7.6. This finding suggests that the original texts were, on average, more readable than the interpretations provided by ChatGPT, with a large effect size (Cohen’s $d = 1.23$), implying a substantial difference in readability between the two text sets. However, variability in the readability difference was observed across the different companies, indicating that the effect of the interpretation process might vary depending on the textual complexity or the specific content of the original documents.

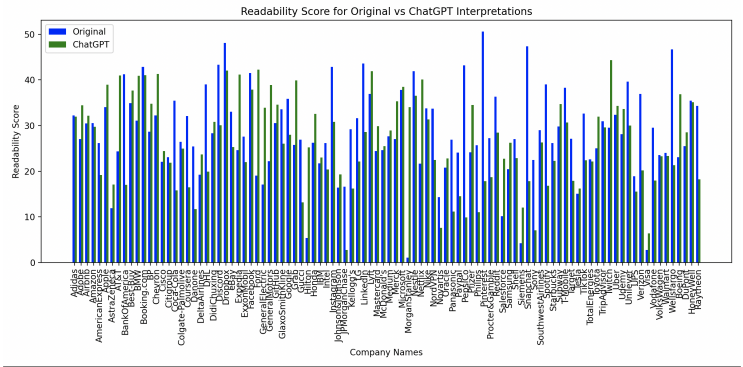


Fig. 1. Readability Scores of Original Privacy Policies vs. ChatGPT Interpretations.

Conclusion: We report on our analysis of the impact of generative AI, specifically ChatGPT-4, on the readability of 102 privacy policies from various industries. Our study reveals a dual-edged performance: while ChatGPT enhanced the readability of 56 policies, it also diminished readability in others. This dichotomy underscores the inherent variability and challenges in automating the interpretation of complex legal texts. The study prompts a deeper investigation into how one might effectively craft prompts for optimal summarization. The feasibility of average users generating "ideal" prompts without specialized knowledge emerges as a question requiring further exploration. These findings underscore the need for advancements in AI models and training techniques that can adeptly navigate the subtleties of intricate documents. Recognizing and addressing the current limitations of generative AI is critical as we seek to incorporate these tools into everyday tasks.

REFERENCES

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[3] Nico Ebert, Kurt Alexander Ackermann, and Björn Scheppler. 2021. Bolder is better: Raising user awareness through salient and concise privacy notices. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–12.

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APPENDIX

List of Organization: Table The privacy policies we assessed were from the following industries: food, fashion, technology, logistics, education, entertainment, oil, gas, journalism, education, air-lines, aerospace, consumer goods, electricity, pharma, luxury, hospitality, banking, e-commerce, social media, automobile, and phone service providers. Here is the comprehensive list of the 102 companies from which we took privacy policies for this study: Adidas, Adobe, Airbnb, Amazon, American Express, Apple, AstraZeneca, AT&T, Bank Of America, Best-Buy, BMW, Booking.com, BP, Chevron, Cisco, Citigroup, Coca-Cola, Colgate-Palmolive, Coursera, Danone, Delta Airlines, DHL, Didi Chuxing, Discord, Dropbox, eBay, Expedia, Exxon Mobil, Facebook, Ford, General Electric, General Motors, GitHub, GlaxoSmithKline, Google, Grab, Gucci, Hilton, Honda, IBM, Intel, Instagram, Johnson&Johnson, JPMorgan Chase, Kellogg’s, LG, LinkedIn, Lyft, Mastercard, McDonald’s, Medium, Merck, Microsoft, Morgan Stanley, Nestle, Netflix, Nike, NordVPN, Novartis, Oracle, Panasonic, Paypal, PepsiCo, Pfizer, Philips, Pinterest, Procter&Gamble, Reddit, Salesforce, Samsung, Shell, Siemens, Snapchat, Sony, Southwest Airlines, Spotify, Starbucks, Subway, T-Mobile, Target, Tesla, TikTok, Total Energies, Toyota, TripAdvisor, Twitch, Uber, Udemy, Unilever, UPS, Verizon, Visa, Vodafone, Volkswagen, Walmart, Wells Fargo, Boeing, DuPont, HoneyWell, and Raytheon. When it came to the actual prompts input into ChatGPT-4 we had to consider that most privacy policies were longer than 25,000 words, so many of the privacy policies input had to be sent in multiple messages. The first message was always, “I will give you a new privacy policy in multiple messages. Can you interpret in a readable format?” followed by the first 25,000 words of the policy. If the policy needed more than 2 messages due to its’ length, we would input the prompt, “Here is more of the same privacy policy: “followed by the next 25,000 words of the policy. This signaled continuity to ChatGPT-4 to have the most continuous interpretation possible. When we reached the end of the policy, we signaled to ChatGPT-4 that this was the end of this policy by inputting, “Here is the last part of the privacy policy: “followed by the rest of the privacy policy. This helped ChatGPT-4 not only wrap up the current interpretation, but also get ready for a new policy which we would signal with the first prompt.

Figures:

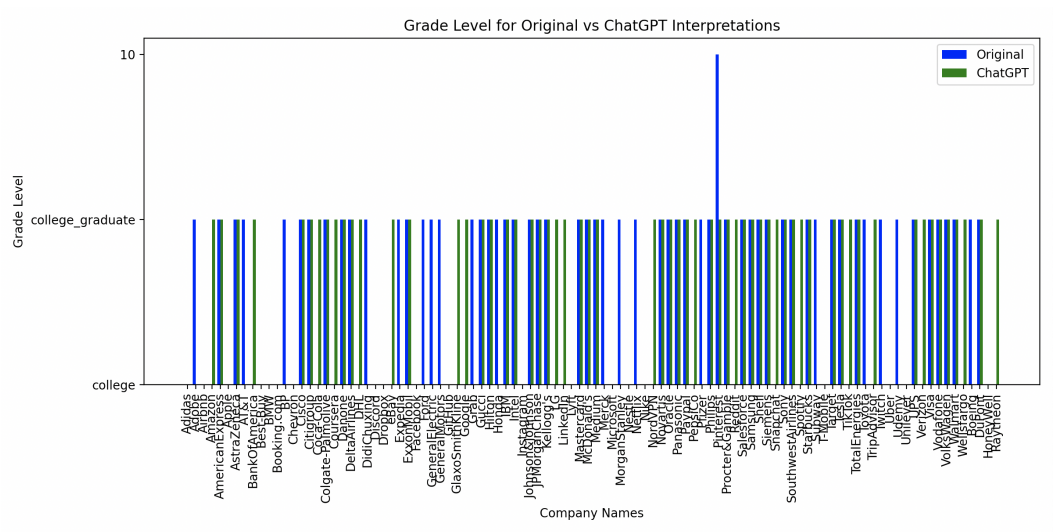


Fig. 2. Which Grade Level Could Read the Original Privacy Policies vs. the ChatGPT Interpretation.

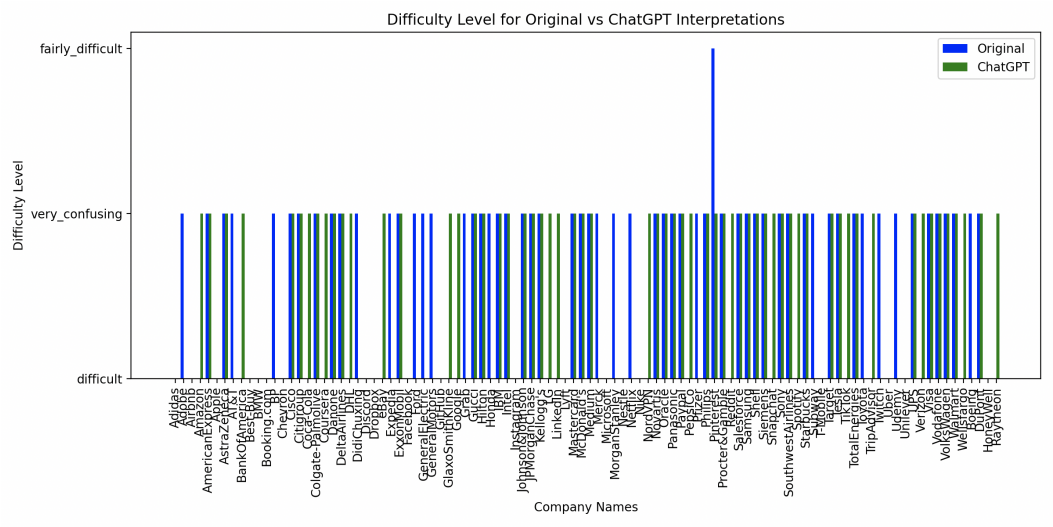


Fig. 3. How Difficult it is to Read the Original Privacy Policies vs. the ChatGPT Interpretation.